

# TLA+ Study Group — Week 03

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## Lecture objective

### What stood out

- in TLA+ every value is a set.
- TLA+ does not specify what set elements are and cannot evaluate expressions like  $42 \in ABC$ .
- The scope of  $\forall$  and  $\exists$  extends as far as possible in TLA+, unless ended by parentheses or the end of a list item such as  $\wedge$  or  $\vee$ .

### Questions

- During video: Why this program does not deadlock. It seems it always should.
- After: Why do we turn off TLC deadlock detection?

### Lecture scope, but not in syllabus

- Precedence graph<sup>1</sup>

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<sup>1</sup>[https://en.wikipedia.org/wiki/Precedence\\_graph](https://en.wikipedia.org/wiki/Precedence_graph)